

4. (a) The photograph shows a fitness smartwatch.



Bethan used a smartwatch to investigate the resting heart rates of ten students in her class.

- Five girls and five boys were chosen at random.
- Each student sat at rest for one minute.
- Then the heart rate shown on the smartwatch was recorded.

The results are shown in **Table 4.1**.

**Table 4.1**

| Girls    |                                          | Boys    |                                          |
|----------|------------------------------------------|---------|------------------------------------------|
| Name     | Resting heart rate<br>(beats per minute) | Name    | Resting heart rate<br>(beats per minute) |
| Seren    | 69                                       | Dan     | 62                                       |
| Katya    | 74                                       | Jim     | 65                                       |
| Nia      | 73                                       | Ifor    | 67                                       |
| Tracy    | 59                                       | Rhys    | 60                                       |
| Angharad | 70                                       | Mohamed | 63                                       |
|          | mean = 69                                |         | mean = .....                             |



Examiner  
only

(i) I. Calculate the mean resting heart rate for the boys.  
**Give your answer to the nearest whole number. Write your answer in Table 4.1.** [3]  
Space for working.

II. State the conclusion that can be made from a comparison of the two means. [1]

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(ii) State **one** way that this investigation is a fair test. [1]

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(b) **Table 4.2** shows the mean resting heart rates by age in women and men from thousands of fitness smartwatch users of many nations and ethnicities.

**Table 4.2**

| Age (years) | Mean resting heart rate<br>(beats per min) |      |
|-------------|--------------------------------------------|------|
|             | Women                                      | Men  |
| 20          | 67.0                                       | 62.5 |
| 30          | 67.5                                       | 63.5 |
| 40          | 68.0                                       | 64.0 |
| 50          | 67.0                                       | 64.5 |
| 60          | 66.0                                       | 64.0 |
| 70          | 65.0                                       | 62.0 |
| 80          | 64.0                                       | 61.0 |

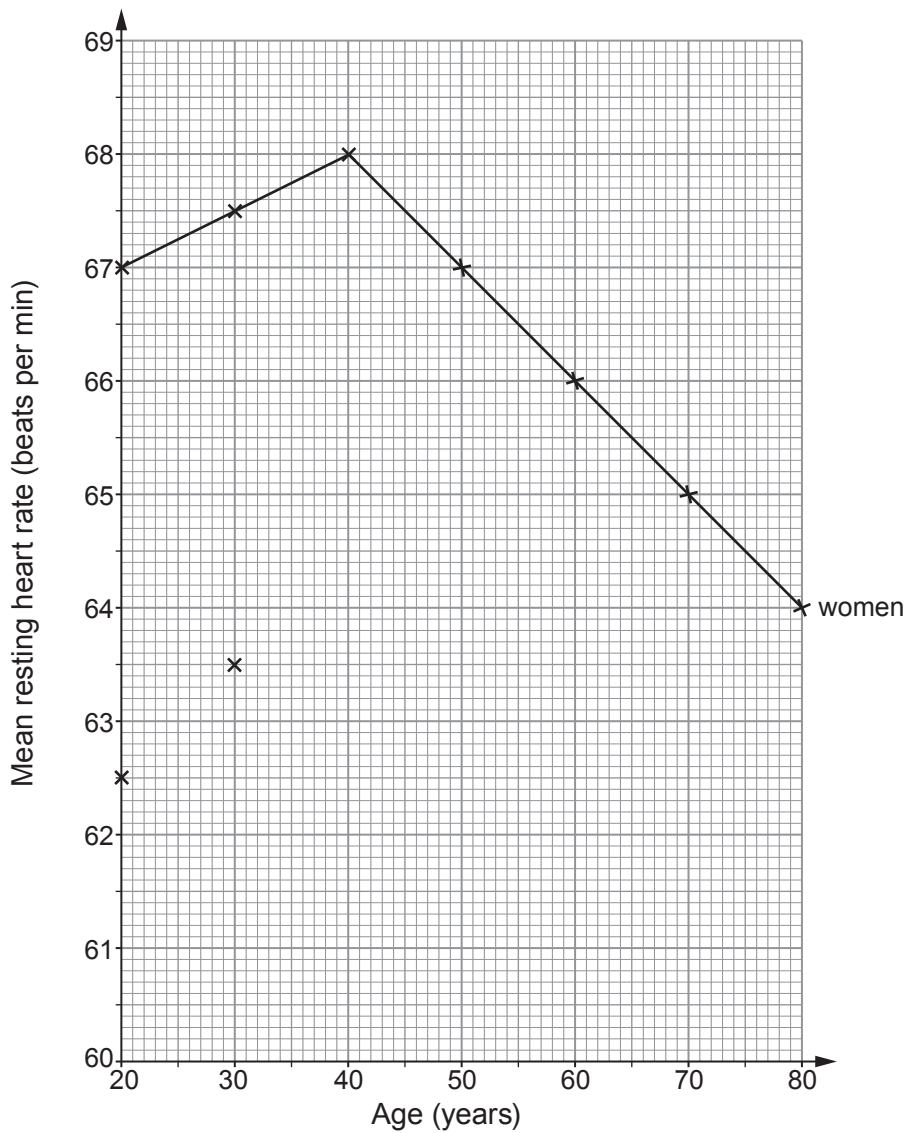
(i) **Graph 4.3** shows the plotted data for women and two plots for men. Complete **Graph 4.3** by:

[3]

- I. plotting the remaining points for the **men** on the grid.
- II. joining **all** the plots **for men** with a ruler.



Graph 4.3



(ii) From **Graph 4.3**:  
State whether the data support the results of Bethan's investigation and explain your answer. [1]

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(iii) Describe how the mean resting heart rate in **men** changes between the ages of 20 and 80 years. [2]

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